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TITLE: BEAUTYTRY: A VIRTUAL COSMETIC TRY-ON MOBILE APPLICATION

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ABSTRACT

Selecting suitable cosmetic products has become a common challenge among consumers especially when purchasing makeup from cosmetics retail stores. Many customers rely on physical cosmetic testers to evaluate product shades before making a purchase. However, the shared use of cosmetic testers raises hygiene concerns, as they may be contaminated with bacteria and viruses, increasing the risk of skin irritation and infection. In addition, some customers feel hesitant to use testers due to cleanliness issues. While others may purchase unsuitable products because they cannot confidently determine the right shade.

To address this issue, our group proposes BeautyTry: a Virtual Cosmetic Try-On Mobile Application that enables users to virtually try on cosmetic products using our prototype product camera. The application allows users to preview different shades of lipsticks products on their faces in real time without direct physical contact like testers. It also provides product information and shades recommendations based on the users facial features and skin tone. The expected outcome of this project is to provide a safer, more hygienic and convenient shopping experience while helping customers make more confident purchasing decisions. Furthermore, the application can reduce dependence on physical cosmetic testers, improve customer satisfaction and support digital transformation in the beauty retail industry.

CHAPTER 1

1.1 INTRODUCTION

The cosmetics business has had trouble providing its customers with an opportunity to purchase a lipstick shade that will complement their skin colour and lip shape. Retailers like MuniKasumi Beauty House have been challenged with how to help their customers find the perfect shade of lipstick. Many traditional methods for selecting lipstick products involve the use of product testing samples by customers. Not only does this raise many sanitary issues, but it may also pose some serious health risks for those who are sampling. In addition to these problems, when consumers try out different shades of lipstick, they are usually able to find one that is completely incorrect for them. This creates a situation where the customer is unhappy with the shade of lipstick they chose and has wasted money buying it. The result is that the customer loses faith in the retailer and becomes less loyal to the brand.

With the above in mind, this research calls for the development of the BeautyTry: A Virtual Cosmetic Try-On Mobile Application an innovative smart device combining HD camera, full-spectrum LED light, and AI technologies to determine lip colour and texture accurately. This smart device deviates from the traditional lipstick shade testing method through emphasizing on non-contact technique which offers more safety and convenience to consumers. In addition, the interactive display will be helpful in offering real-time recommendations and virtual try-on of lipsticks shades, thus enabling customers to make the best decision on the desired product.

This research, accordingly seeks to explore the success of using the BeautyTry: Virtual Cosmetic Try-On Mobile Application in enhancing the consumer's experience, making fewer mistakes in the process of selecting products and building the brands credibility in cosmetic retailers like MuniKasumi Beauty House. Solving the problem of proper choice of lipstick shades, removing the issues of hygiene during product testing and preventing consumers from buying wrong lipstick shades, this research aims at developing personalized beauty technologies and responding to the increasing needs for smart and sustainable retailing solutions.

1.2 PROJECT BACKGROUND

The beauty and cosmetics industry have grown rapidly as consumers increasingly rely on online platforms to purchase beauty products. However, one of the main challenges faced by customers when buying cosmetics online is the difficulty in visualizing how a product, especially makeup shades such as lipstick and foundation will look on their own face. This issue may cause uncertainty, dissatisfaction and product returns due to unsuitable choices.

To overcome this problem, BeautyTry is developed as a virtual beauty try-on application that utilizes Augmented Reality (AR) technology to allow users to try cosmetic products digitally before making a purchase. The application provides a more interactive and convenient shopping experience by helping users visualize suitable products based on their appearance.

This project aims to improve customer confidence in selecting beauty products while supporting cosmetic retailers in providing a modern and engaging digital shopping experience by helping customers make more confident purchasing decisions while reducing dependence on shared cosmetic testers. Additionally, BeautyTry supports cosmetics retailers by providing a more engaging and modern approach to showcase their products to customers.

1.3 PROBLEM STATEMENT

1. Difficulty choosing the right lip matte shade

- Many customers find it difficult to choose the most suitable lip matte shade when shopping online because they cannot test the product before purchasing.

2. Risk of purchasing the wrong product

- Customers may receive a lip matte shade that does not match their skin tone or personal preference, leading to dissatisfaction and unnecessary spending.

3. Hygiene concerns when testing lip matte products

- Many customers are concerned about hygiene when using lip matte testers in physical stores because the same product may be used by multiple people. This increases the risk of bacteria transfer and discourages customers from trying cosmetic products.

1.4 RESEARCH OBJECTIVE

1. To develop a virtual lip matte try-on application.
2. To provide a hygienic way to try lip matte products.
3. To reduce the risk of purchasing the wrong lip matte shade

1.5 RESEARCH QUESTIONS

1. How can a virtual lip matte try-on application help customers choose the right shade?
2. How can the application provide a hygienic way to try lip matte products?
3. How can the application reduce the risk of purchasing the wrong lip matte shade?

1.6 PROJECT SCOPE

This research focuses on the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application, a prototype application designed to help users identify the most suitable lipstick shades based on their facial features and skin tone using artificial intelligence. The study emphasizes the use of AI-based image analysis to provide personalized lipstick recommendations through an easy-to-use digital platform.

The scope of this research is limited to the development and testing of the prototype application. The system is designed to analyze users' facial images and recommend lipstick shades from a predefined database. Although the study primarily targets individuals interested in cosmetics and beauty products, the findings are intended to demonstrate the potential application of AI technology in the beauty industry, particularly in enhancing customer shopping experiences and personalized beauty recommendations.

1.7 IMPORTANCE OF STUDY

The significance of this study lies in its contribution to both the beauty industry and the advancement of artificial intelligence technology. BeautyTry: Virtual Cosmetic Try-On Mobile Application provides users with a convenient and efficient way to discover lipstick shades that complement their skin tone and facial appearance, reducing the uncertainty often associated with purchasing cosmetic products.

Furthermore, this study demonstrates how artificial intelligence can be integrated into the cosmetics industry to improve customer satisfaction and support informed purchasing decisions. The project also serves as an educational platform by showcasing the practical implementation of AI, image processing, and software development in solving real-world problems. By promoting personalized beauty recommendations through digital technology, this study contributes to innovation and the digital transformation of the beauty industry.

1.8 OPERATIONAL FINDINGS

Operational findings from this research show that the virtual cosmetic try-on mobile application performs perfectly well in ensuring a smart and clean cosmetic product retail environment. Integration of HD smart cameras along with the full spectrum LED lights helps the device in capturing lip colour and texture data accurately, which is then processed by the AI system for recommending suitable lipstick shades. In the process of testing, the system proved its capability to decrease customer doubts through real-time results with high accuracy in terms of matching. Moreover, the system also helped in engaging customers through the virtual try-on feature, as it allowed them to see the effect of various lipstick shades on their lips without applying any product physically. This also helped in overcoming the hygiene issue associated with the traditional methods.

1.9 EXPECTED FINDINGS

This project is expected to help customers choose the most suitable lip matte shade through an AI-powered virtual try-on feature. The application also provides a hygienic alternative to physical cosmetic testers, allowing customers to try different shades safely and confidently before making a purchase while reducing the risk of buying the wrong product.

Otherwise, this project is expected to improve the overall online cosmetic shopping experience by increasing customers' confidence in selecting lip matte products. It may also benefit cosmetic retailers by enhancing customer satisfaction, reducing incorrect purchases, and encouraging more informed purchasing decisions.

1.10 SUMMARY

The above chapter has highlighted the basis for this study, which includes the background, problem statement, objectives, and scope of the study. This study is significant since it addresses actual problems being faced by cosmetics retailers and introduces innovation by incorporating the use of artificial intelligence in customized beauty solutions. Operational findings indicate that there is potential in the BeautyTry: A Virtual Cosmetic Try-On Mobile Application to enhance hygiene, accuracy and satisfaction when choosing products. In the following chapter, there will be a review of relevant literature in order to provide theoretical justification for this proposed solution.

CHAPTER 2 (LITERATURE RESEARCH)

2.1 INTRODUCTION

A literature review is an essential component of any research project as it establishes the theoretical and empirical foundation of the study. By reviewing previous studies, researchers can identify existing knowledge, research gaps, and relevant theories that support the development of new solutions. In the context of this research, the literature review examines previous studies related to virtual cosmetic try-on technologies, artificial intelligence (AI), augmented reality (AR), mobile applications, and consumer behaviour in the cosmetics retail industry. This ensures that the proposed BeautyTry: A Virtual Cosmetic Try-On Mobile Application is supported by evidence-based findings and current technological advancements.

The purpose of the literature review is also to provide a strong academic basis for understanding the challenges faced by cosmetic retailers and consumers when selecting suitable lipstick products. Previous studies have highlighted issues such as inaccurate shade matching, hygiene concerns associated with product testers, and customer dissatisfaction resulting from unsuitable cosmetic purchases. By exploring these findings, researchers can better justify the need for a virtual cosmetic try-on solution that improves shopping convenience while enhancing customer confidence and satisfaction.

Another important role of the literature review is to examine the application of emerging technologies in the beauty retail industry. Recent advancements in artificial intelligence, augmented reality, image processing, and facial recognition have enabled consumers to virtually test cosmetic products in real time without physical contact. Reviewing these technologies allows researchers to evaluate their effectiveness in improving shade matching accuracy, reducing product selection errors, and enhancing the overall customer experience. Furthermore, these technological innovations support retailers in strengthening brand credibility, increasing customer engagement, and promoting more sustainable retail practices by minimizing product waste.

Overall, the literature review provides a comprehensive understanding of the concepts, theories, and technological developments related to virtual cosmetic applications. The knowledge gained from previous studies supports the development of BeautyTry as an innovative mobile application that addresses hygiene concerns, improves lipstick shade selection accuracy, enhances customer satisfaction, and contributes to the growing adoption of smart digital solutions in the cosmetics retail industry.

2.2 PREVIOUS STUDY

Virtual Cosmetic Applications

Augmented Reality (AR) and Virtual Reality (VR) technologies can improve consumer engagement, decision-making, and satisfaction by allowing users to visualize cosmetic outcomes before undergoing procedures. However, the study also emphasized that these technologies require careful implementation to ensure realistic expectations and protect users' personal information. These findings demonstrate the potential of AR and VR applications in providing more interactive and personalized experiences within the cosmetic industry. (Haykal et al., 2024)

Found that Augmented Reality (AR)-based cosmetic try-on applications can improve consumer product evaluation and purchase decisions. The study highlighted that factors such as informativeness, ease of use, and enjoyment influence users' trust and intention to use AR cosmetic applications. The findings suggest that AR technology can enhance consumer engagement and satisfaction by providing a more interactive and informative shopping experience. (Micheletto et al., 2025)

A virtual cosmetics try-on system provides a realistic try-on experience for consumers and helps them efficiently choose suitable cosmetics. In this article, we propose a real-time augmented reality virtual cosmetics try-on system for smartphones (ARCosmetics), taking speed, accuracy, and stability into consideration at each step to ensure a better user experience. A novel and very fast face tracking method utilizes the face detection box and the average position of facial landmarks to estimate the faces in continuous frames. (An et al., 2023)

Health Risks of Testers

Investigated microbial contamination in lip cosmetic testers and found that some tester products exceeded safety limits for harmful microorganisms. The study highlighted that repeated exposure to multiple users may increase hygiene risks in cosmetic retail environments. Therefore, non-contact technologies such as virtual cosmetic try-on applications can serve as an alternative solution to reduce contamination risks while improving consumer safety. (Kim et al., 2025)

Cosmetic testers are widely used in retail settings to allow customers to try products before purchase. However, their frequent exposure to multiple users raises significant concerns regarding microbial contamination. Improper handling and repeated contact can turn these products into potential reservoirs of pathogenic microorganisms, posing health risks ranging from minor skin infections to severe systemic illnesses, particularly in individuals with compromised immunity. (Shaikh & Kataria, 2025)

Makeup testers at beauty counters could expose users to harmful viruses, bacteria and fungi, warned a dermatologist, urging the public to avoid applying products directly to the lips, face or around the eyes. Aesthetic dermatologist Dr Lim Ing Kien, also known as Dr Ingky on social media, said makeup testers can harbour viruses such as herpes simplex virus type 1 (HSV-1) as well as bacteria and fungi that may cause skin infections, eye conditions and allergic reactions. (The Sun, 2025)

Cosmetic testers are widely used in retail settings to allow customers to try products before purchase. However, their frequent exposure to multiple users raises significant concerns regarding microbial contamination. Improper handling and repeated contact can turn these products into potential reservoirs of pathogenic microorganisms, posing health risks ranging from minor skin infections to severe systemic illnesses, particularly in individuals with compromised immunity. (Shaikh & Kataria, 2025)

Shared cosmetic testers, particularly lipsticks and lip balms, may increase the risk of herpes simplex virus (HSV-1) transmission. This highlights the importance of adopting safer and more hygienic alternatives, such as virtual cosmetic try-on applications, to minimize direct contact with cosmetic products. (Kosmo, 2024)

Risks of Inaccurate Shade Matching in Cosmetics

The virtual makeup can also be utilized at offline stores to prevent possible sanitation problems associated with swatching. Faithful color reproduction is one of the most important factors in virtual makeup applications. Thus, the color difference between the virtual and real makeup results needs to be minimized. However, most previous studies on virtual makeup focus on the recommendation of makeup style rather than on the accuracy of color reproduction. (Kim & Lee. 2020)

A virtual cosmetics try-on system provides a realistic try-on experience for consumers and helps them efficiently choose suitable cosmetics. In this article, we propose a real-time augmented reality virtual cosmetics try-on system for smartphones (ARCosmetics), taking speed, accuracy, and stability into consideration at each step to ensure a better user experience. (An et al., 2023)

Virtual make-up try-on applications utilizing augmented reality (AR) technology have been introduced to allow users to virtually test cosmetics before purchasing. Features such as shade-matching tools enable the app to analyze the user's facial image and recommend suitable cosmetic shades. This technology helps reduce user uncertainty when selecting cosmetic colors, particularly during online shopping.(Ameen et al. 2022)

Technical factors, such as the lighting quality of facial images, can influence the accuracy of beauty product recommendation systems. Suboptimal lighting conditions can lead to inaccurate skin tone analysis, subsequently resulting in mismatched cosmetic color recommendations.(Afshar et al. 2023)

2.3 SUMMARY

The literature review highlights that traditional cosmetic purchasing practices, particularly physical product testing and manual shade selection, present several challenges for both consumers and cosmetic retailers. Previous studies have identified that the use of cosmetic testers may increase the risk of microbial contamination and hygiene concerns, while inaccurate shade matching can lead to customer dissatisfaction, unsuitable purchases, and reduced confidence towards cosmetic brands.

Furthermore, previous research has demonstrated that the integration of Artificial Intelligence (AI), Augmented Reality (AR), and virtual try-on technologies can provide effective solutions by enabling consumers to visualize cosmetic products virtually before making purchasing decisions. These technologies support more accurate shade recommendations, improve consumer engagement, enhance shopping convenience, and reduce the need for direct product testing.

Based on the findings from previous studies, the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application is justified as a potential solution to overcome existing limitations in cosmetic retail. By providing a non-contact virtual try-on experience with accurate shade visualization and personalized recommendations, BeautyTry can improve customer satisfaction, minimize product selection errors, address hygiene concerns, and support the adoption of smarter and more sustainable digital solutions within the cosmetics industry.

CHAPTER 3 (METHODOLOGY/PROJECT DESIGN)

3.1 INTRODUCTION

This chapter elaborate more about methodology and design for our project BeautyTry : A Virtual Cosmetic Try-On Mobile Application. This methodology meets the needs and expectations of users in selecting suitable lip matte products. This application aims to provide a more convenient and interactive experience by allowing users to visualize different lip matte shades before making a purchase decision.

To ensure the application is developed effectively. This systematic methodology is implemented throughout the project development process. Methodology consists of four main stages which is problem identification, data collection, design development and design evaluation. Problem identification is conducted to understand the challenges faced by users, while data collection focuses on gathering relevant information and user requirements. The design development stage involves creating the application based on the identified needs, followed by design evaluation to determine the usability and effectiveness of the application.

Through this approach, the project aims to produce an application that can improve user confidence in choosing lip matte products while providing a safer and more efficient alternative to traditional cosmetic testing methods.

3.2 STUDY DESIGN

This design study focuses on the development of a BeautyTry: A Virtual Cosmetic Try-On Mobile Application that aims to assist users in selecting suitable lipstick shades based on their skin tone and facial features through a digital platform. The system design incorporates Artificial Intelligence (AI) technology by using a camera, facial scanning system, and image analysis to detect the user's facial features, especially the lip area, accurately.

To support the implementation of this application during an event or product demonstration, the system is designed with several main components, including a face detection camera, proper lighting equipment to illuminate the user's face, and a tablet screen that displays the application interface and real-time lipstick shade simulation results. The lighting system helps the camera capture clearer facial images, while the facial scanning technology enables the application to analyze facial characteristics and apply suitable lipstick shades digitally.

Furthermore, the design of this application considers important aspects such as user-friendly interaction, accurate colour detection, and an enhanced virtual try-on experience. Through the integration of hardware and software components, users can explore and select lipstick shades confidently without the need for physical product testers. This approach helps to improve hygiene practices, reduce the risk of product contamination, and provide a more convenient and engaging shopping experience within the beauty industry.

3.2.1 SAMPLING

The sampling method used in this study was purposive sampling, whereby respondents were selected based on specific criteria that aligned with the objectives of the research. A total of 30 respondents participated in the questionnaire survey. The respondents consisted of individuals who had experience purchasing or using cosmetic products, particularly lip products such as lipstick, lip tint, and lip matte. They were chosen because they represent the target users of the proposed system, BeautyTry: A Virtual Cosmetic Try-On Mobile Application

The questionnaire focused on respondents' experiences and challenges when selecting lipstick shades, including issues such as inaccurate colour representation on product packaging, unsuitable lighting conditions in cosmetic stores, the unavailability of product testers, and concerns regarding the hygiene of shared testers. In addition, respondents were asked to provide their opinions on whether an Artificial Intelligence (AI), based on system could help reduce colour selection errors, minimise product wastage and improve the overall shopping experience.

The data collected from the 30 respondents provided valuable insights into users needs, preferences, and acceptance of AI technology in cosmetic shade selection. These findings served as an important reference for the design and development of the BeautyTry: A Virtual Cosmetic Try-On Mobile Application, ensuring that the proposed system meets user requirements and addresses the challenges identified through the survey.

3.2.2 DATA COLLECTION METHODS

The first method involved distributing questionnaires to collect information from respondents regarding their experiences in selecting the appropriate lip matte shade, the risk of purchasing an unsuitable product, and hygiene concerns associated with testing lip matte products. The structured format of the questionnaire allowed the data to be collected consistently, making the responses easier to analyse and providing valuable insights into customer preferences and experiences.

In addition to the questionnaires, direct observation was carried out at Muni Kasumi Beauty House. During the observation, it was found that not all lip matte products were provided with tester units. As a result, customers often faced difficulties in identifying their desired shade before making a purchase, which could reduce customer satisfaction and increase the likelihood of buying an unsuitable product. Furthermore, the availability of tester products created additional challenges for the retailer, as the tester stock was frequently depleted and required replacement. Hygiene issues were also observed due to multiple customers using the same tester products.

The findings from the direct observation were consistent with the questionnaire results, strengthening the conclusion that customers experience challenges in selecting the correct lip matte shade, while retailers at MuniKasumi Beauty House face significant issues related to tester availability, stock management, and hygiene.

3.2.3 DATA ANALYSIS METHODS

The data collected from the questionnaire were analyzed using quantitative methods. Descriptive statistics, including frequencies and percentages, were used to analyze the respondents' opinions and experiences regarding lipstick shade selection and the need for the proposed BeautyTry application. The results were presented in the form of charts and graphs to facilitate data interpretation.

Based on the analysis, 90.5% of the respondents identified lipstick as the most difficult cosmetic product to purchase online. In addition, 50% of the respondents stated that they had purchased a cosmetic product whose actual shade differed from their expectations after application, while 21.4% reported that this issue occurred frequently. These findings indicate that many consumers experience difficulties in selecting the appropriate lipstick shade.

The findings also revealed that 64.3% of the respondents believed that different lighting conditions in stores were the main factor affecting the appearance of lipstick shades, followed by 23.8% who identified mobile phone screen lighting as another contributing factor. Furthermore, 45.2% of the respondents stated that choosing the wrong lipstick shade resulted in wasted money, while 40.5% reported that they had to purchase another product because the selected shade was unsuitable.

Regarding hygiene concerns, 69% of the respondents strongly agreed and 31% agreed that sharing cosmetic testers among customers could lead to hygiene issues. These findings suggest that consumers prefer a safer and more convenient alternative for trying lipstick shades before making a purchase.

Overall, the data analysis indicates that consumers face significant challenges in selecting suitable lipstick shades and require a more accurate and convenient solution. Therefore, the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application is proposed to assist users in identifying suitable lipstick shades based on their skin tone using Artificial Intelligence (AI). This application is expected to reduce purchasing errors, improve user confidence, and provide a more hygienic alternative to conventional cosmetic testers.

3.3 SUMMARY

This chapter has outlined the methodology and project design adopted in the development of BeautyTry: A Virtual Cosmetic Try-On Mobile Application. The research process began with problem identification, followed by purposive sampling of 30 respondents who represented the target users. Data was collected through questionnaires and secondary sources such as news reports and analyzed using both quantitative and qualitative methods. The findings confirmed that users face significant challenges in shade selection due to inaccurate packaging references, unsuitable store lighting and hygiene concerns with shared testers. Based on these insights, the BeautyTry system was designed to integrate AI technology, facial scanning and real-time shade simulation, providing a user-centered, hygienic and efficient solution. The structured methodology ensured that the application meets user needs while offering a safer and more interactive alternative to traditional cosmetic testing.

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APPENDIX

1. Project Cost Estimation

COMPONENT	QUANTITY	UNIT PRICE (RM)	TOTAL (RM)
IPad 11 th Generation 11inch (RAM 4GB+)	1	871.00	1,371.00
LED Fill Light Bar	2 pieces	22.00	44.00
Adjustable Tablet Holder	1	18.00	18.00
Acrylic Sheet 5mm	1	36.00	36.00
MDF Board / Acrylic Base	1	31.00	31.00
Hollow Steel Frame	4 pieces	18.00	72.00
Skru, Nut and Bolt	1 set	12.00	12.00
Epoxy / Acrylic Glue	1	14.00	14.00
Extension Plug	1	21.00	21.00
Paint / Vinyl Sticker	2	27.00	54.00
Laptop with the AMD Ryzen 3 5300U (14-inch 16GB DDR4 RAM)	1	1,700.00	1,700.00

Total Cost: RM2,853.00 +-

1. iPad 11th Generation (11-inch, RAM 4GB+)

Serves as the main display for the BeautyTry application. It allows users to scan their face, detect their skin tone, and virtually try on different lipstick shades before making a purchase.

2. LED Fill Light Bar (2 pieces)

Provides sufficient and even lighting during the face scanning process, ensuring accurate skin tone detection and improving the quality of the virtual lipstick try-on.

3. Adjustable Tablet Holder

Securely holds the iPad in place and allows the viewing angle to be adjusted for user comfort while ensuring the camera captures the user's face properly.

4. Acrylic Sheet (5mm)

Used to construct the outer body and protective panels of the prototype, giving BeautyTry a clean, modern, and professional appearance.

5. MDF Board / Acrylic Base

Acts as the base of the prototype, providing stability and support for all structural and electronic components.

6. Hollow Steel Frame (4 pieces)

Forms the main framework of the prototype, supporting the iPad, lighting system, and acrylic panels while ensuring the structure is strong and durable.

7. Screws, Nuts, and Bolts (1 set)

Used to assemble and securely fasten all parts of the prototype, ensuring structural stability and safety.

8. Epoxy / Acrylic Glue

Used to bond acrylic panels and other components together, creating a strong, neat, and durable structure.

9. Extension Plug

Supplies electrical power to the iPad charger and LED lights, allowing all electronic components to operate simultaneously.

10. Paint / Vinyl Sticker

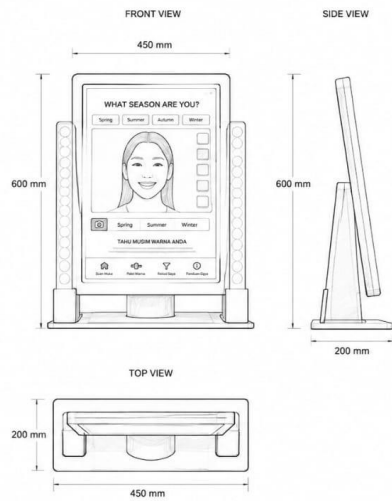
Applied as the finishing touch to enhance the prototype's appearance and display the BeautyTry branding, logo, and user instructions.

11. Laptop with AMD Ryzen 3 5300U (14-inch, 16GB DDR4 RAM)

Used to develop, test, and manage the BeautyTry application, ensuring the system functions properly before being deployed on the iPad.

2. Product Sketch

Sketch The Design



EXPLANATION : BeautyTry : A Virtual Cosmetic Try-On Mobile Application

The prototype design of BeautyTry: A Virtual Cosmetic Try-On Mobile Application was carefully planned to provide users with an interactive, hygienic, and convenient cosmetic testing experience. The prototype consists of a tablet mounted on a stable stand, integrated with a front-facing camera and dual LED light bars. The design enables users to virtually try different lipstick shades in real time without the need for physical cosmetic testers.

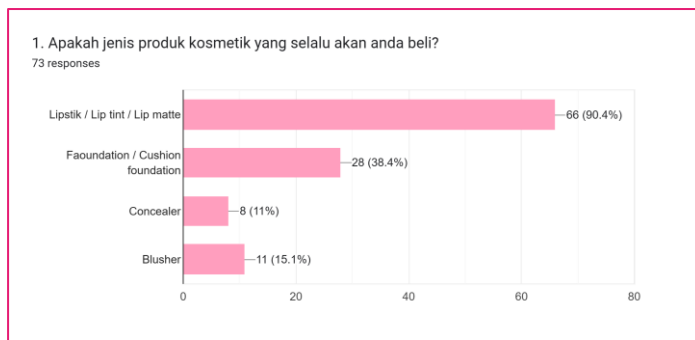
From the front view, the prototype features a tablet that serves as the main user interface. The application displays a live camera feed and provides several lipstick shade options for users to select. Once a shade is chosen, the application uses artificial intelligence (AI) facial detection technology to accurately identify the user's lips and instantly apply the selected lipstick colour on the screen. The interface also includes navigation buttons that allow users to switch between lipstick shades, compare colours, and save their preferred selection.

From the side view, the tablet is positioned at an ergonomic viewing angle to ensure user comfort during the virtual try-on session. Behind the tablet, a support frame securely holds the device in place, while internal cable management keeps the overall appearance neat and organised. Two LED light bars are installed on both sides of the tablet to provide consistent illumination, improving facial detection accuracy and ensuring realistic colour rendering under different lighting conditions.

In terms of hardware components, the prototype includes an Android tablet, a built-in or external HD camera, dual LED fill lights, a tablet holder, and a sturdy base constructed from acrylic or MDF material. The tablet runs the BeautyTry mobile application, while the AI-based facial recognition system processes the user's face and overlays virtual lipstick shades in real time. The LED lighting enhances image quality, allowing the application to produce more precise and natural-looking cosmetic simulations.

The overall structure is designed to be compact, portable, and user-friendly, making it suitable for use in cosmetic stores, shopping malls, beauty exhibitions, and promotional events. By replacing traditional cosmetic testers with virtual technology, the BeautyTry prototype helps improve hygiene, reduces product wastage, and enables customers to make more confident purchasing decisions through an engaging digital experience.

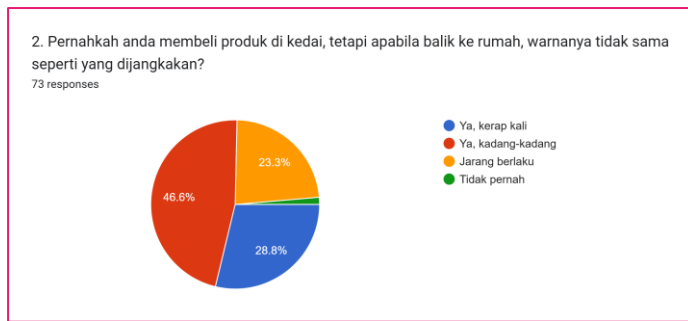
3. Questionnaire Survey



Question 1: What type of cosmetic product will you always buy?

Based on the survey conducted with 73 respondents, 66 respondents (90.4%) stated that they regularly purchase lip products (lipstick, lip tint, or lip matte), making it the most popular cosmetic category. This is followed by foundation/cushion foundation (38.4%), blusher (15.1%), and concealer (11.0%). The findings indicate that lip products are the cosmetic items most frequently purchased by consumers, highlighting their importance in the beauty market.

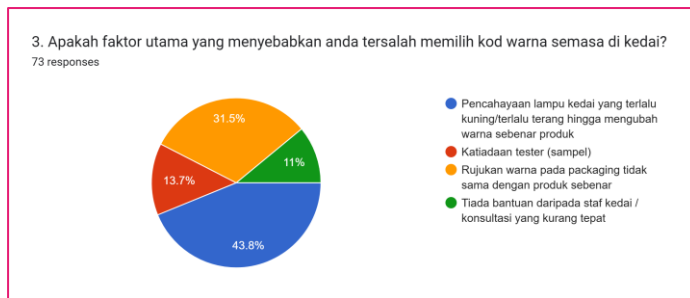
Summary: The survey shows that lip products are the most frequently purchased cosmetic items, with 90.4% of respondents choosing them. This supports the project's focus on lip matte products, as they represent the highest consumer demand and have significant market potential.



Question 2: Have you ever bought a product in a store, but when you got home, the colour wasn't the same as expected?

Based on the survey conducted with 73 respondents, 46.6% of the participants stated that they sometimes experience a discrepancy where the product colour changes upon returning home, while 28.8% reported that this happens frequently. Conversely, 23.3% indicated it rarely occurs, and only a negligible percentage have never faced this issue. These statistics reveal that over 75% of consumers have faced colour inconsistency after purchase, highlighting a widespread issue with physical in-store colour selection.

Summary: The data demonstrates that a vast majority of consumers experience unexpected colour mismatches after purchasing cosmetics in-store. This reinforces the necessity of this project in developing a more reliable colour-matching solution to eliminate post-purchase dissatisfaction.



Question 3: What are the main factors that cause you to choose the wrong colour code while in the store?

Based on the survey conducted with 73 respondents, 43.8% identified improper in-store lighting (too yellow or too bright) as the primary cause for selecting the wrong cosmetic shade. The second major factor is misleading packaging design, where the outer colour reference does not match the actual product (31.5%), followed by the total absence of physical testers (13.7%) and a lack of assistance from store staff or beauty consultants (11.0%). The findings highlight that environmental lighting and inaccurate packaging are major barriers to accurate shade selection.

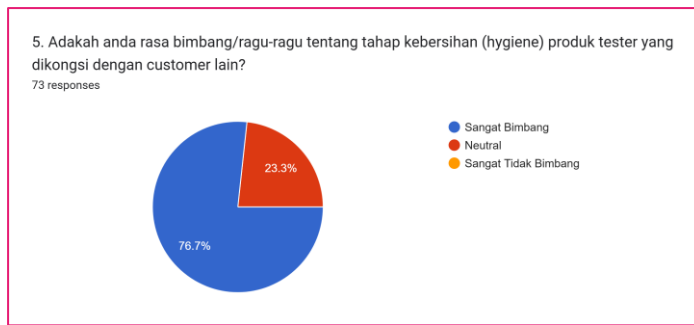
Summary: In-store lighting and misleading packaging are the top external factors causing consumers to choose the wrong cosmetic shades. This validates the project's objective to bypass these unreliable environmental factors by offering a digitalized evaluation process.



Question 4: What are the consequences when you buy the wrong colour code?

Based on the survey conducted with 73 respondents, nearly half of the participants (49.3%) stated that buying the wrong colour code leads to direct financial waste because the product is ultimately left unused. Additionally, 31.5% of respondents are forced to purchase alternative products to mix and adjust the shade to fit their skin tone, while 19.2% continue to wear the mismatched product despite it not suiting their complexion. The results emphasize that incorrect shade selection imposes a notable financial and practical burden on consumers.

Summary: Selecting the wrong colour code results in prominent financial waste and forces consumers to spend more money on corrective products. This highlights a clear consumer pain point that this project aims to solve by preventing costly purchasing mistakes.



Question 5: Are you worried/doubtful about the level of hygiene of the tester products shared with other customers?

Based on the survey conducted with 73 respondents, a resounding majority of 76.7% expressed that they are deeply concerned about the sanitary conditions of shared cosmetic testers in retail stores. The remaining 23.3% maintained a neutral stance, while notably, 0% of the respondents felt completely unconcerned. This data clearly underscores an intense, widespread anxiety among modern consumers regarding bacterial contamination and hygiene risks tied to public tester products.

Summary: An overwhelming majority of consumers harbour severe concerns regarding the hygiene and safety of shared physical testers. This strong anxiety justifies the urgent need for a touchless, virtual testing alternative, which is a core value proposition of this project.



Question 6: If a cosmetics store uses AI, do you think it will reduce the risk of waste and hygiene?

Based on the survey conducted with 72 respondents, 34.7% strongly believe that integrating Artificial Intelligence (AI) would be highly effective in minimizing product waste and hygiene risks, while an additional 26.4% agree that it would offer some level of assistance. Meanwhile, 26.4% remain uncertain, and only a small minority of 12.5% feel it would not help. With over 61% of respondents leaning towards a positive outlook, the data signals robust consumer readiness and optimism toward AI adoption in the beauty retail sector.

Summary: The majority of consumers view the implementation of AI as a viable and helpful solution to combat hygiene issues and product waste. This positive consumer reception strongly validates the development and relevance of the AI-driven system proposed in this FYP.

4. Gantt Chart

